

Yousef Kart

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SKILLS

- **Languages & Tools:** C++, Python, Java, TypeScript, SQL
- **Tools:** Git, CMake, ImGui, Qt, Vercel, Docker, FastAPI, Supabase, OpenGL, Three.js, Vitest, Playwright
- **Technologies:** React, React Native, Next.js, Node.js, Expo, Tailwind, PostgreSQL, REST APIs

EXPERIENCE

Full Stack Software Developer – Red Piston – Windsor, ON Apr 2026 – Present

- Built and deployed a workforce-management platform using Next.js, Supabase, and Vercel, consolidating 3 applications into one with authentication, subscriptions, RBAC, applicant tracking, and AI-powered support
- Rebuilt a legacy Android application using React Native, Expo, and Strapi, expanding support to iOS and web, reducing load times, and providing a real-time dashboard for clients
- Developed a new multi-object 3D rendering feature for an AR platform with 1K+ users

Software Engineer – Scelta – Windsor, ON Feb 2025 – Sept 2025

- Led the design and rollout of a shared UI component library used across several production apps, removing hundreds of duplicate UI files and accelerating feature delivery
- Built an SSO system enabling seamless authentication across 3 applications, eliminating redundant login flows
- Developed a real-time 3D geospatial visualization pipeline for interactive development mapping tools
- Reduced REDMap's initial loading time by 97% via query optimization and endpoint restructuring
- Designed a production-ready Next.js application template that decreased new project build time by more than 50%
- Mentored a team of 6 Master of Computer Science students, defining project requirements, conducting regular progress reviews, evaluating performance, and ensuring deliverables were completed on time

3D Modeler & Designer – Freelance – Remote June 2021 – Feb 2025

- Delivered 3D assets and rendering pipelines for 40+ global clients, optimizing polygon counts and texture workflows to meet real-time performance constraints
- Produced architectural visualizations for \$100M+ developments while reducing asset turnaround times by 66%

PROJECTS

ABG Horizon (Website) – Python, FastAPI, Next.js, Supabase, LLMs, Finnhub, Massive, SEC EDGAR, yfinance

- Engineered an event-driven Python pipeline that ingests real-time financial news, performs LLM-based event classification and sentiment analysis, and generates ranked trading signals using liquidity, relative volume, volatility, and historical metrics

Monte Carlo Simulation Engine (GitHub) – C++20, OpenGL, GLFW, ImGui, CMake

- Developed a high-performance C++ Monte Carlo simulation engine using Geometric Brownian Motion (GBM) with OpenGL rendering and Dear ImGui controls for real-time stochastic asset price visualization

Black-Scholes Model Visualizer (GitHub) – C++17, Qt, CMake

- Developed a C++ Black-Scholes pricing engine implementing analytical Greeks (Delta, Gamma, Vega, Theta, Rho) with real-time visualization and interactive parameter analysis using Qt

Flag Frenzy (Roblox) – Roblox Studio, Lua, Blender

- Built and launched a multiplayer Roblox shooter with custom gameplay systems and optimized 3D content, scaling to 2.5M monthly visits before being acquired for \$50K USD

EDUCATION

University of Windsor – 4x Dean's Honour Roll

Sept 2021 – May 2025

Bachelor of Computer Science (Honours) & Minor in Mathematics